Short-time asymptotics for solutions of the evolutionary game theoretic p-laplacian and Pucci operators

Diego Berti

Instituto de Ciencias Matemáticas (ICMAT), Madrid

12th International ISAAC Congress
Geometric & Regularity Properties of Solutions to Elliptic and Parabolic PDEs
University of Aveiro, 29 July – 2 August 2019

Let $u = u(x,t)$ be the solution to the following initial-boundary value problem:

$$
\begin{aligned}
u_t - F\left(\nabla u, \nabla^2 u\right) &= 0 \qquad \text{in } \ \Omega \times (0,\infty), && (1)\\
u &= 0 \qquad \text{on } \ \Omega \times \{0\}, && (2)\\
u &= 1 \qquad \text{on } \ \Gamma \times (0,\infty). && (3)
\end{aligned}
$$

where Ω is a domain in $\mathbb{R}^N$, with $N \geq 2$, with (not necessarily bounded) boundary $\Gamma \neq \varnothing$.

Here, F is a generic operator that will be specified in what follows. Since we shall consider nonlinear operators not in divergence form, we shall intend that (1) is satisfied in the sense of the **theory of viscosity solutions**.

We shall focus on the **short-time behavior** of u.

The linear case – Varadhan formula

Theorem ([Varadhan, 1967])

Let u be the solution to $u_t - \Delta u = 0$ in $\Omega \times (0, \infty)$, satisfying (2) and (3). Then, we have that

$$\lim_{t \to 0^+} 4t \log u(x,t) = -d_\Gamma(x)^2 \quad \text{for any} \quad x \in \overline{\Omega}. \tag{4}$$

Here, for $x \in \Omega$, $d_\Gamma(x)$ is the distance of x from the boundary Γ:

$$d_\Gamma(x) = \inf\{|y - x| : y \in \Gamma\}.$$

Formula (4) holds when one merely asks that $\Gamma = \partial\left(\mathbb{R}^N \setminus \overline{\Omega}\right)$ and it is satisfied uniformly on every compact subset of $\overline{\Omega}$.

The linear case – Formula for the heat content in balls touching Γ

A more intimate link between short-time diffusion and the geometry of the domain Ω is shown by the following theorem:

Theorem ([Magnanini, Sakaguchi, *Proc Roy. Soc. Edinburgh*, 2007])

Assume $\Omega \in C^2$. Let $x \in \Omega$ and suppose that the open ball $B_R(x)$ centered at x and with radius R is contained in Ω and such that $\overline{B_R(x)} \cap \Gamma = \{y_x\}$ for some $y_x \in \Gamma$. Then we have:

$$\lim_{t \to 0^+} t^{-\frac{N+1}{4}} \int_{B_R(x)} u(z,t)\, dz = C_N \left(\prod_{j=1}^{N-1} \left[\frac{1}{R} - \kappa_j(y_x) \right] \right)^{-\frac{1}{2}},$$

where $\kappa_1, \dots, \kappa_{N-1}$ are the principal curvatures of Γ at y_x and C_N is a positive constant.

The linear case – Formula for the heat content

The formula holds true even if $k_j(y_x) = \frac{1}{R}$ for some $j = 1, \dots, N-1$ and it is satisfied if the domain is smooth only in a neighborhood of y_x.

Formula for the mean value

Simply by multiplying both sides of the previous formula by $\frac{R^{\frac{N+1}{2}}}{|B_R|}$, we get the following formula for the mean value on $B_R(x)$:

$$\lim_{t\to 0^+} \left(\frac{R^2}{t}\right)^{\frac{N+1}{4}} \fint_{B_R(x)} u(z,t)\,dz = C_N \left(\prod_{j=1}^{N-1} \left[1 - R\kappa_j(y_x)\right]\right)^{-\frac{1}{2}}, \tag{5}$$

where C_N is a positive constant.

Game-Theoretic p-Laplacian

The **game-theoretic p-laplacian** is defined for $1 < p < \infty$ by

$$\Delta_p^G u = \frac{1}{p} |\nabla u|^{2-p} \Delta_p u$$

where Δ_p is the usual p-laplacian

$$\Delta_p u = \mathrm{div}\left(|\nabla u|^{p-2} \nabla u\right)$$

and in the extremal case $p = \infty$ by

$$\Delta_\infty^G u = \left(\nabla^2 u \frac{\nabla u}{|\nabla u|} \cdot \frac{\nabla u}{|\nabla u|} \right).$$

Note that, for any $1 < p \leq \infty$ we formally have that

$$\Delta_p^G u = \frac{\Delta u}{p} + \left(1 - \frac{2}{p}\right) \Delta_\infty^G u.$$

Pucci operators

Let $0 < \lambda \leq \Lambda < \infty$, the **Pucci operators** $\mathcal{M}^{\pm}$ are defined by:

$$\mathcal{M}^{+}\left(\nabla^2 u\right) = \lambda \sum_{e_i<0} e_i + \Lambda \sum_{e_i>0} e_i$$

and

$$\mathcal{M}^{-}\left(\nabla^2 u\right) = \Lambda \sum_{e_i<0} e_i + \lambda \sum_{e_i>0} e_i$$

where e_i, for $i = 1, \ldots, N$, are the eigenvalues of $\nabla^2 u$.

Pucci operators

Let $0 < \lambda \leq \Lambda < \infty$, the **Pucci operators** $\mathcal{M}^{\pm}$ are defined by:

$$\mathcal{M}^{+}\left(\nabla^2 u\right) = \lambda \sum_{e_i<0} e_i + \Lambda \sum_{e_i>0} e_i$$

and

$$\mathcal{M}^{-}\left(\nabla^2 u\right) = \Lambda \sum_{e_i<0} e_i + \lambda \sum_{e_i>0} e_i$$

where e_i, for $i = 1, \ldots, N$, are the eigenvalues of $\nabla^2 u$.

Equivalently, $\mathcal{M}^{\pm}$ can be seen as the following extremal operators:

$$\mathcal{M}^{+}\left(\nabla^2 u\right) = \sup\{\mathrm{tr}\left(A\nabla^2 u\right) : \ A \in \mathcal{A}_{\lambda,\Lambda}\}$$

and

$$\mathcal{M}^{-}\left(\nabla^2 u\right) = \inf\{\mathrm{tr}\left(A\nabla^2 u\right) : \ A \in \mathcal{A}_{\lambda,\Lambda}\}$$

where $\mathcal{A}_{\lambda,\Lambda}$ is the subspace of symmetric matrices whose eigenvalues belong to $[\lambda, \Lambda]$.

Our contributions

In collaboration with Magnanini, we have generalized Formulas (4) and (5) to the cases of nonlinear diffusion associated to either Δ_p^G or $\mathcal{M}^{\pm}$.
Moreover, we have extended these results showing sharp estimates of the convergence in (4) depending on the regularity of the domain and new asymptotic formulas for some statistical quantities called q-means (which we shall define later).

These results are contained in the following works:

- [B.-Magnanini, *J. Math. Pures Appl.*, 2019];
- [B. (Ph.D. thesis) 2019];
- [B.-Magnanini (in preparation)].

Similar asymptotic results for a family of elliptic problems for Δ_p^G are contained in

- [B.-Magnanini, *Applicable Analysis*, 2019].

To unify the presentation of the results, we introduce the parameter α depending on the operator taken into consideration.

Definition of α

$$\alpha = \begin{cases} p' & \text{when } F = \Delta_p^G \\ 1/\Lambda & \text{when } F = \mathcal{M}^+ \\ 1/\lambda & \text{when } F = \mathcal{M}^- \end{cases}.$$

Here, that given $p \in (1, \infty]$,

$$p' = \begin{cases} \frac{p}{p-1} & \text{if } \; p \in (1, \infty) \\ 1 & \text{se } \; p = \infty. \end{cases}$$

Varadhan-type formulas

We are ready to state the analog of (4).

Theorem ([B.-Magnanini, JMPA], [B.-Magnanini, in preparation])

Let Ω be a domain such that $\Gamma = \partial\left(\mathbb{R}^N \setminus \overline{\Omega}\right)$. Let u be the solution to (1)-(3)*. Then, it holds that*

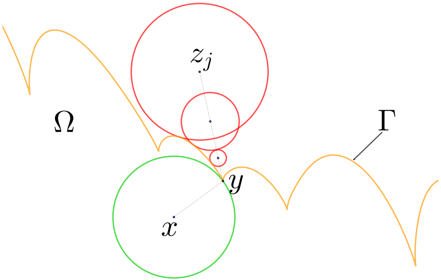
$$-\lim_{t\to 0^+} 4t \log u(x,t) = \alpha\, d_\Gamma(x)^2 \quad x \in \overline{\Omega}. \tag{6}$$

Remark

Observe that the parameter α merely appears as a multiplying constant in the r.h.s. of the formula.

The following is a (brief) scheme on which Formula (6) relies:

– For any $x \in \Omega$ let $y \in \Gamma$ be a point that realizes $|x-y| = d_\Gamma(x)$. Since the assumption $\Gamma = \partial\left(\mathbb{R}^N \setminus \overline{\Omega}\right)$ holds, a sequence of points $z_j \in \mathbb{R}^N \setminus \overline{\Omega}$ approaching y exists.



– Since comparison and maximum principles hold, an analysis on radial cases both in a ball or in the complement of a ball produces both a barrier from above and a sequence of barriers from below, one for each z_j.

– Accurate estimates on the asymptotic profile of the barriers and a limit argument as $|z_j - y| \to 0$ give (6).

To construct the barriers we consider radial solutions, for which both Δ_p^G and $\mathcal{M}^\pm$ take a rather simple form. How does this help?

- In the ball, we bound u from above by involving $u^\varepsilon = u^\varepsilon(x)$ the solution to

$$\begin{aligned} u^\varepsilon - \varepsilon^2 F\left(\nabla u^\varepsilon, \nabla^2 u^\varepsilon\right) = 0 \qquad & \text{in } \ \Omega \\ u^\varepsilon = 1 \qquad & \text{on } \ \Gamma, \end{aligned}$$

  where, $\varepsilon > 0$. Since a proper relation between u and u^ε holds, we take advantage of the quite explicit expression of u^ε in terms of *modified Bessel functions* (in the case Ω is a ball).
- In the complement of a ball, we essentially make use of the existence of a (radial) global solution of (1).

Asymptotics for q-means on $B_R(x)$

Given $x \in \Omega$, $t > 0$ and $q \in [1, \infty]$, the ***q*-mean** $\mu_q(x,t)$ on $B_R(x) \subset \Omega$ of u the solution to (1)-(3) is the unique real number μ such that

$$\|u(\cdot, t) - \mu\|_{L^q(B_R(x))} \leq \|u(\cdot, t) - \lambda\|_{L^q(B_R(x))} \text{ for any } \lambda \in \mathbb{R}.$$

Remark

μ_q generalizes the **mean value** of u on $B_R(x)$, which is obtained when one considers $q = 2$ in the previous definition:

$$\mu_2(x,t) = \fint_{B_R(x)} u(z,t)\, dz.$$

Asymptotics for q-means

Theorem ([B.-Magnanini, JMPA], [B.-Magnanini, in prep.])

Let $x \in \Omega$ and $R > 0$ such that $\overline{B_R(x)} \cap \Gamma = \{y_x\}$ for some $y_x \in \Gamma$. Let u be the solution to (1)-(3). Given $q \in (1, \infty]$, let $\mu_q(x,t)$ the q-mean of u on $B_R(x)$. Then, for any $q \in (1, \infty)$, we have that

$$\lim_{t \to 0^+} \left(\frac{R^2}{t} \right)^{\frac{N+1}{4(q-1)}} \mu_q(x,t) = C_{N,q} \left(\alpha^{\frac{N+1}{2}} \prod_{j=1}^{N-1} \left[1 - R\,\kappa_j(y_x) \right] \right)^{-\frac{1}{2(q-1)}}, \qquad (7)$$

where $C_{N,q}$ is a positive constant depending only on the labeled parameters.

Notice the role of α and q.

In the extremal case $q = \infty$, we have obtained that

$$\mu_\infty(x,t) \to \frac{1}{2} \text{ per } t \to 0^+.$$

Key points on which the proof of (7) is based are:

- The behavior for small values of $s > 0$ of

$$\mathcal{H}_{N-1}\left(\Gamma_s \cap B_R(x)\right)$$

  in terms of the principal curvatures. Here, $\Gamma_s = \{x \in \Omega : d_\Gamma(x) = s\}$, for $s > 0$ and $\mathcal{H}_{N-1}$ is the $(N-1)$-dimensional Hausdorff measure.
- Improvement of the barriers in the case of C^2 domain from uniform sharp estimates (new also in the linear case) on the rate of convergence of the Varadhan-type formulas (6).

Uniform sharp estimates

Suppose that Ω is of class $C^{0,\omega}$, i.e. Γ is locally a graph of a continuous function whose modulus of continuity is controlled by ω. We have the following result.

Theorem (Uniform sharp estimates [B.-Magnanini, JMPA],[B.-Magnanini, in preparation])

Let u be the solution to (1)-(3). Then, we have that

$$4\, t \log u(x,t) + \alpha\, d_\Gamma(x)^2 = O\left(t\, \log \psi_\omega(t)\right),$$

for $t \to 0^+$, uniformly on compact subsets of $\overline{\Omega}$.

- The function $\psi_\omega(t)$ is positive and (depending on ω) vanishes for $t \to 0^+$.
- If $\Omega \in C^{0,\beta}$, with $0 < \beta < 1$, then

  $$O\left(t \log \psi_\omega(t)\right) = O\left(t \log t\right),$$

  for $t \to 0^+$, uniformly on compact subsets of $\overline{\Omega}$.

Elliptic case

Consider now the solution u^ε to the elliptic problem

$$\begin{aligned} u^\varepsilon - \varepsilon^2 F\left(\nabla u^\varepsilon, \nabla^2 u^\varepsilon\right) = 0 \qquad & \text{in} \quad \Omega \\ u^\varepsilon = 1 \qquad & \text{on} \quad \Gamma, \end{aligned}$$

for $\varepsilon > 0$.
We are able to give the **asymptotic profile of** u^ε **for small** $\varepsilon \to 0^+$.

Theorem ([B.-Magnanini, AA], [B.-Magnanini, in preparation])

Assume that $\Gamma = \partial\left(\mathbb{R}^N \setminus \overline{\Omega}\right)$. *Then, it holds that*

$$\lim_{\varepsilon \to 0^+} \varepsilon \log u^\varepsilon(x) = -\sqrt{\alpha}\, d_\Gamma(x), \ \ x \in \overline{\Omega}.$$

Elliptic uniform estimates – GT p-laplacian

Theorem ([B.-Magnanini, AA])

Let $\Omega \subset \mathbb{R}^N$ and u^ε be the solution to $u^\varepsilon - \varepsilon^2 \Delta_p^G u^\varepsilon = 0$ in Ω such that $u^\varepsilon = 1$ on Γ. If Ω is such that $\Gamma = \partial\left(\mathbb{R}^N \setminus \overline{\Omega}\right)$, then we have that:

$$\varepsilon \log u^\varepsilon + \sqrt{p'}\, d_\Gamma = \begin{cases} O(\varepsilon) & \text{if } \ p = \infty, \\ O(\varepsilon \log \varepsilon) & \text{if } \ p > N, \end{cases}$$

for $\varepsilon \to 0^+$, on every compact subset of $\overline{\Omega}$.
If $\Omega \in C^{0,\omega}$, it holds that

$$\varepsilon \log u^\varepsilon + \sqrt{p'}\, d_\Gamma = \begin{cases} O(\varepsilon \log |\log \psi_\omega(\varepsilon)|) & \text{if } \ p = N, \\ O(\varepsilon \log \psi_\omega(\varepsilon)) & \text{if } \ 1 < p < N, \end{cases}$$

for $\varepsilon \to 0^+$, uniformly on every compact subset of $\overline{\Omega}$.

Elliptic uniform estimates – Pucci operators

Theorem ([B.-Magnanini, in preparation])

Assume $\Omega \in C^{0,\omega}$. *Let* u^ε *be the solution to* $u^\varepsilon - \varepsilon^2 \mathcal{M}^- \left(\nabla^2 u^\varepsilon\right) = 0$ *in* Ω *such that* $u^\varepsilon = 1$ *on* Γ. *Then, it holds that for* $\varepsilon \to 0^+$ *uniformly on compact subsets of* $\overline{\Omega}$:

$$\varepsilon \log u^\varepsilon + \frac{d_\Gamma}{\sqrt{\lambda}} = \begin{cases} O\left(\varepsilon \log|\log \psi_\omega(\varepsilon)|\right) & \textit{if} \ \ N = 2 \ \ \textit{and} \ \ \lambda = \Lambda, \\ O\left(\varepsilon \log \psi_\omega(\varepsilon)\right) & \textit{if} \ \ N \neq 2 \ \ \textit{or} \ \ \lambda \neq \Lambda. \end{cases}$$

Let u^ε *be the solution to* $u^\varepsilon - \varepsilon^2 \mathcal{M}^+ \left(\nabla^2 u^\varepsilon\right) = 0$ *in* Ω *such that* $u^\varepsilon = 1$ *on* Γ. *Then, it holds that for* $\varepsilon \to 0^+$ *uniformly on compact subsets of* $\overline{\Omega}$:

$$\varepsilon \log u^\varepsilon + \frac{d_\Gamma}{\sqrt{\Lambda}} = \begin{cases} O\left(\varepsilon \log \varepsilon\right) & \textit{if} \ \ \Lambda > \lambda(N-1), \\ O\left(\varepsilon \log|\log \psi_\omega(\varepsilon)|\right) & \textit{if} \ \ \Lambda = \lambda(N-1), \\ O\left(\varepsilon \log \psi_\omega(\varepsilon)\right) & \textit{if} \ \ \Lambda < \lambda(N-1). \end{cases}$$

Thanks for your kind attention!